10.3 Density Problems

Density links geometry to real-world materials: $\rho = \dfrac{m}{V}$ (mass per unit volume), so $m = \rho V$ and $V = \dfrac{m}{\rho}$ — the same three-way relationship as distance/rate/time. A separate but related idea, **population density**, is people per unit area: $D = \dfrac{P}{A}$. Most density problems here are two-step: first use a volume formula from this unit to find V (or $B \cdot h$), *then* apply the density relationship. This worksheet also covers comparing densities (which object is denser, or would float in water, since water ≈ 1 g/cm^3), weighted-average density of composite objects, and unit conversions.

Examples

Example 1: Density from mass and volume

A solid has mass 240 g and volume 30 cm^3. Find its density.

Solution

Identify the formula: density is mass per unit volume, $\rho = \dfrac{m}{V}$.

Compute: $\rho = \dfrac{240}{30} = 8$.

Answer: 8 g/cm^3.

Example 2: Mass from density (cube)

A cube of edge 4 cm has density 2.5 g/cm^3. Find its mass.

Solution

First find the volume: $V = s^3 = 4^3 = 64$ cm^3.

Then $m = \rho V = 2.5 \times 64 = 160$ g.

Answer: 160 g.

Example 3: Volume from mass and density

An object has mass 90 g and density 3 g/cm^3. Find its volume.

Solution

Identify: rearrange $\rho = \dfrac{m}{V}$ to solve for volume instead of density: $V = \dfrac{m}{\rho}$.

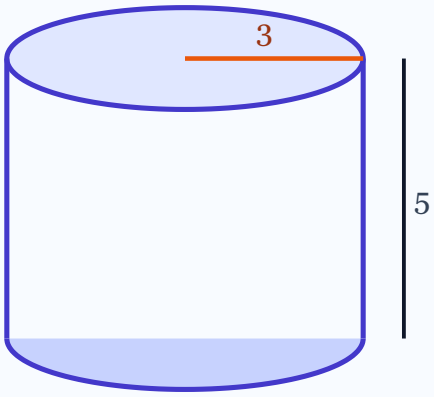
Compute: $V = \dfrac{90}{3} = 30$.

Answer: 30 cm^3.

Example 4: Cylinder — find volume, then density

A metal cylinder has radius 3 cm, height 5 cm, and mass 450π g. Find its density.

Solution



Find the cylinder's volume first, then divide mass by volume.

Step 1 — volume: $V = \pi r^2 h = \pi(9)(5) = 45\pi$ cm^3.

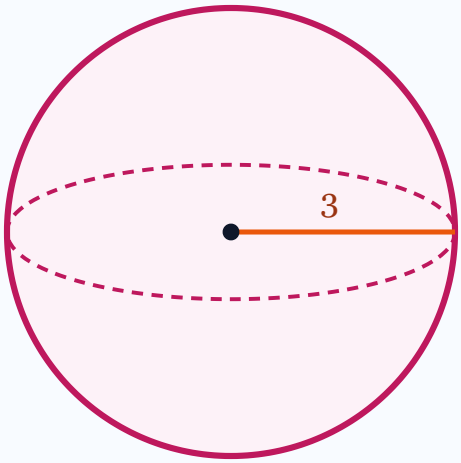
Step 2 — density: $\rho = \dfrac{m}{V} = \dfrac{450\pi}{45\pi}$. The π cancels: $\rho = 10$ g/cm^3.

Answer: 10 g/cm^3.

Example 5: Sphere — find volume, then density

A metal ball bearing (sphere) has radius 3 cm and mass 288π g. Find its density.

Solution



Sphere volume first, then $\rho = m/V$.

Step 1 — volume: $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi \text{ cm}^3$.

Step 2 — density: $\rho = \frac{m}{V} = \frac{288\pi}{36\pi} = 8 \text{ g/cm}^3$.

Answer: 8 g/cm^3.

Example 6: Does it float? (compare to water)

A block has mass 540 g and volume 600 cm^3. Find its density and determine whether it floats in water (water has density 1 g/cm^3).

Solution

Compute the density: $\rho = \frac{540}{600} = 0.9 \text{ g/cm}^3$.

Compare to water: since $0.9 < 1$, the block is less dense than water, so it **floats**.

Answer: $\rho = 0.9 \text{ g/cm}^3$; it floats.

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Example 7: Population density

A town of $48{,}000$ people covers 16 km^2. Find the population density.

Solution

Identify the formula: population density is people per unit area, $D = \dfrac{P}{A}$.

Compute: $D = \dfrac{48{,}000}{16} = 3000$.

Answer: 3000 people per km^2.

Example 8: Population from area and density

A region has area 25 mi^2 and population density 200 people/mi^2. Find the population.

Solution

Identify: rearrange $D = \dfrac{P}{A}$ to solve for population instead of density: $P = D \times A$.

Compute: $P = 200 \times 25 = 5000$.

Answer: 5000 people.

Example 9: Area from population and density

A city has population 9000 and population density 300 people/km^2. Find its area.

Solution

Identify: rearrange $D = \dfrac{P}{A}$ to solve for area instead: $A = \dfrac{P}{D}$.

Compute: $A = \dfrac{9000}{300} = 30$.

Answer: 30 km^2.

Example 10: Composite object — weighted-average density

An object is cast from two materials fused together: Part 1 has volume $20\ \text{cm}^3$ and density $8\ \text{g/cm}^3$; Part 2 has volume $30\ \text{cm}^3$ and density $3\ \text{g/cm}^3$. Find the object's overall (average) density.

Solution

Step 1 — mass of each part: $m_1 = \rho_1 V_1 = 8(20) = 160\ \text{g}$; $m_2 = \rho_2 V_2 = 3(30) = 90\ \text{g}$.

Step 2 — totals: total mass $= 160 + 90 = 250\ \text{g}$; total volume $= 20 + 30 = 50\ \text{cm}^3$.

Step 3 — overall density: $\rho_{\text{avg}} = \dfrac{250}{50} = 5\ \text{g/cm}^3$.

Note: the average density is **not** simply $\dfrac{8+3}{2} = 5.5$ — it must be weighted by volume, since the two parts don't have equal volume in general (here it happened close, but always compute total mass over total volume).

Answer: $5\ \text{g/cm}^3$.

Example 11: Unit conversion

Aluminum has density $2.7\ \text{g/cm}^3$. Convert this to kg/m^3.

Solution

Convert the units piece by piece: $1\ \text{g} = 0.001\ \text{kg}$, and $1\ \text{m}^3 = (100\ \text{cm})^3 = 1{,}000{,}000\ \text{cm}^3$.

$$2.7\ \frac{\text{g}}{\text{cm}^3} \times \frac{0.001\ \text{kg}}{1\ \text{g}} \times \frac{1{,}000{,}000\ \text{cm}^3}{1\ \text{m}^3} = 2.7 \times 1000 = 2700\ \text{kg/m}^3.$$

Shortcut: $1\ \text{g/cm}^3 = 1000\ \text{kg/m}^3$ always, so just multiply by 1000.

Answer: $2700\ \text{kg/m}^3$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Density

Mass 150 g, volume 25 cm^3. Find density.

Answer: **Verification:**

Problem 2: Mass (cube)

Cube edge 3 cm, density 5 g/cm^3. Find mass.

Answer: **Verification:**

Problem 3: Volume

Mass 72 g, density 8 g/cm^3. Find volume.

Answer: **Verification:**

Problem 4: Cylinder mass

Cylinder radius 2, height 7, density 3 g/cm^3. Find the volume, then the mass (leave π).

Answer: **Verification:**

Problem 5: Population density

$12{,}000$ people on 8 km^2. Find the population density.

Answer: **Verification:**

Problem 6: Population

Density 400 people/mi^2, area 15 mi^2. Find the population.

Answer: **Verification:**

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Problem 7: Sphere mass

Sphere radius 2 cm, density 6 g/cm^3. Find the volume, then the mass (leave π).

Answer: **Verification:**

Problem 8: Compare

Object A: 40 g in 8 cm^3. Object B: 45 g in 9 cm^3. Which is denser?

Answer: **Verification:**

Problem 9: Cone mass

Cone radius 3, height 4, density 2 g/cm^3. Find the volume, then the mass (leave π).

Answer: **Verification:**

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Problem 10: Area

Population 9000, density 300 people/km^2. Find the area.

Answer: **Verification:**

Problem 11: Density

Mass 200 g, volume 50 cm^3. Find density.

Answer: **Verification:**

Problem 12: Prism density

Rectangular prism $4 \times 5 \times 6$ cm, mass 360 g. Find its density.

Answer: **Verification:**

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Problem 13: Formula

Rearrange $\rho = \frac{m}{V}$ to solve for m.

Answer: **Verification:**

Problem 14: Float or sink?

A wood block has mass 350 g, volume 500 cm^3. A metal block has mass 800 g, volume 100 cm^3. Find both densities and state which float(s) in water (1 g/cm^3).

Answer: **Verification:**

Problem 15: Challenge

A statue is cast from bronze (volume 500 cm^3, density 8.9 g/cm^3) fused to a gold-plated cap (volume 20 cm^3, density 19.3 g/cm^3). Find the total mass and the statue's overall density.

Answer: **Verification:**

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✓ Answer Key

1. 6 g/cm^3
2. 135 g
3. 9 cm^3
4. $V = 28\pi$; $m = 84\pi$ g
5. 1500 people/km^2
6. 6000
7. $V = \dfrac{32\pi}{3}$; $m = 64\pi$ g
8. Neither — both are 5 g/cm^3
9. $V = 12\pi$; $m = 24\pi$ g
10. 30 km^2
11. 4 g/cm^3
12. 3 g/cm^3
13. $m = \rho V$
14. Wood: 0.7 g/cm^3, floats; metal: 8 g/cm^3, sinks
15. Total mass 4836 g; overall density 9.3 g/cm^3

Great work — keep practising!